TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Fabric Crease Recovery

How completely a sharply folded fabric springs back when the creasing load is removed — the Shirley test of crease recovery, wrinkle resistance and easy-care performance.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 8 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the crease-recovery angle of a woven fabric in the warp and weft directions, on face and back, by the Shirley crease-recovery method (ISO 2313 — determination of the recovery from creasing of a folded specimen, measured in the horizontal plane), and to express the result as mean recovery angles, total crease recovery and percentage recovery, with an interpretation for garment appearance.

1.1 What creasing does to a fabric

A crease is a sharp, localised bend. When fabric is folded, fibres on the outside of the fold are stretched, fibres on the inside are compressed, and yarns are forced to slide and flatten against each other at the fold line. If the deformation is wholly elastic, the fold disappears the instant the load is removed. In practice part of the deformation is plastic — held by newly formed hydrogen bonds in cellulose, by molecular slip in thermoplastics, or by yarns locking in their displaced positions — and the fabric opens only part-way, leaving a residual crease angle. Crease recovery measures the balance between elastic return and plastic set. Good recovery means a garment that sheds wrinkles in wear and after laundering; poor recovery means a garment that must be ironed to look acceptable.

The test therefore isolates one simple quantity: after a specimen is folded exactly in half, creased under a standard load for a standard time, and then allowed to recover for a standard time, the angle between its two arms is measured. A flat, uncreased specimen lies at 180°; a perfectly retained fold lies at 0°. Real fabrics fall between the two, and the larger the angle, the better the recovery.

$$\text{Total crease recovery (°)} = \text{warp recovery angle} + \text{weft recovery angle} \quad (\text{maximum possible} = 360°)$$

$$\text{Percentage recovery (\%)} = \text{total recovery} / 360 \times 100$$

$$\text{Crease retention (°)} = 360° - \text{total recovery}$$

Warp and weft are always reported separately as well as summed. The two directions rarely crease alike: warp yarns are usually finer, more highly twisted and more densely set than weft yarns, and they have survived different tensions in weaving and finishing. Face and back can also differ where a finish, coating or pile lies mainly on one side, so a complete result tests both directions and states which face was folded outward.

1.2 Why fibres differ

Recovery is first a fibre property. Wool recovers superbly because its keratin structure — coiled alpha-helices cross-linked by disulfide bonds — acts as a three-dimensional spring: it resists bending and pulls the fibre back when released. Polyester recovers well for a different reason, a stiff, resilient, hydrophobic molecular chain with little moisture-assisted bond re-formation. Cotton and viscose recover poorly when dry-bent and worse when wet, because water breaks the inter-chain hydrogen bonds that hold cellulose in shape, the chains slip under the creasing stress, and new hydrogen bonds then form in the creased position — the fabric has, in effect, been re-shaped. This is precisely why resin-based easy-care finishes work on cotton: cross-links introduced into the cellulose pin the chains so they cannot slip and re-bond in the folded state, trading a little strength and abrasion resistance for a large gain in recovery. Linen, with its highly crystalline, stiff cellulose, creases sharply and recovers worst of the common fibres — the crisp wrinkle of linen is a material fact, not a quality failure.

1.3 The Shirley principle

The Shirley crease-recovery tester standardises every variable that would otherwise make angles incomparable: specimen size, fold geometry, creasing load, creasing time, recovery time and the geometry of measurement. The specimen, a rectangular strip, is folded in half with the fold pressed between glass plates under a defined load — conventionally a 2 kg load in the Shirley instrument — for 5 minutes. The load sets the severity of the crease in a reproducible way, standing in for the sharp, sustained fold of a trouser crease or a packed garment. The folded specimen is then transferred, without handling the fold, to the tester: one arm is clamped horizontally and the other hangs free, and after 5 minutes of free recovery the angle of the free arm is read against a graduated circular scale. The two 5-minute periods matter: recovery is time-dependent, rapid in the first seconds and then slowing, so an angle read at 1 minute and an angle read at 5 minutes are different results on the same fabric. All specimens are first conditioned in the standard testing atmosphere of 65% relative humidity and 20 °C, because regain changes both the stiffness of the fibres and the ease with which hydrogen bonds break and reform.

Like all single-number appearance tests, crease recovery has limits that the student should state in any report. It measures recovery from one sharp fold in a strip; it does not directly measure the rumpled,

multi-directional wrinkling of a worn garment, for which panel rating against photographic replicas is used commercially. It is, however, precise, cheap, repeatable and highly sensitive to fibre type and finish — which is why it remains the laboratory standard for comparing fabrics and for controlling easy-care finishing.

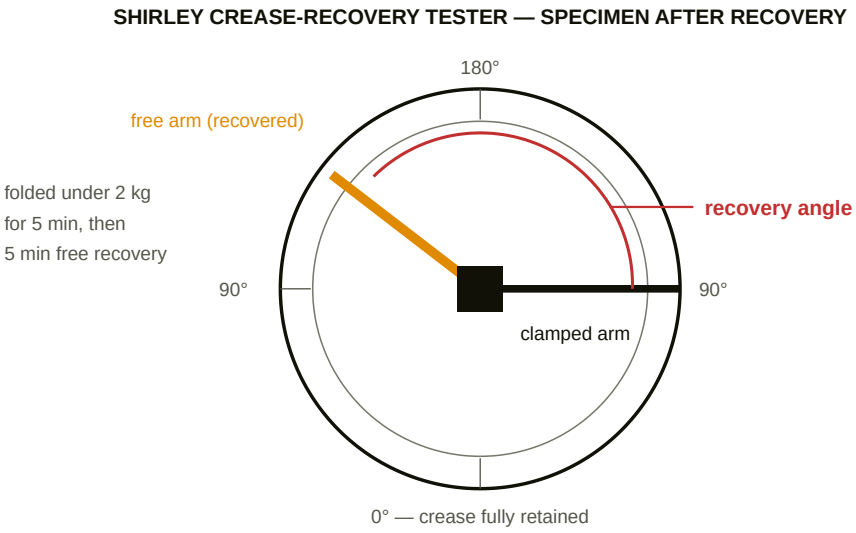


Figure 1 — The Shirley crease-recovery tester. One arm of the previously folded specimen is clamped horizontally; the free arm rises as the crease opens, and its angle against the graduated dial — 0° for a fully retained fold, 180° for complete recovery — is read after 5 minutes.

2. Apparatus and procedure

- **Apparatus:** Shirley crease-recovery tester with graduated circular scale and specimen clamp; glass plates and standard creasing load (2 kg); stopwatch; cutting template and scissors; tweezers for handling specimens without touching the fold; conditioning facilities for the standard atmosphere (65% RH, 20 °C).
- **Specimens:** cut rectangular strips to the size specified for the instrument (commonly about 40 mm × 15 mm in the Shirley method), clear of selvedges, creases and faults — at least four specimens in the warp direction and four in the weft direction for this exercise, with the long dimension parallel to the direction under test so that the fold crosses that direction. Mark the face of each specimen.
- **Conditioning:** condition all specimens in the standard atmosphere (65% RH, 20 °C) until in equilibrium, and carry out the test in the same atmosphere.
- **Creasing:** fold each specimen exactly in half, face outward (or as specified and recorded), place the fold between the glass plates, and apply the standard load smoothly for exactly 5 minutes. Do not iron, rub or pre-form the fold by hand.
- **Measurement:** at the end of 5 minutes remove the load, lift the folded specimen with tweezers by one arm only, and clamp that arm horizontally in the tester so the fold sits at the centre of the dial

and the free arm hangs freely. Allow exactly 5 minutes of free recovery, then read the angle of the free arm on the graduated scale to the nearest degree. The specimen must not touch the scale, the bench or the operator during recovery.

- **Replication and reporting:** repeat for all warp and weft specimens (and repeat with the back outward if a full characterisation is required). Report the mean recovery angle for each direction, the total recovery (warp + weft), the percentage recovery, and the test conditions. Note any specimen that twisted instead of opening in its own plane — twisted recovery is recorded, not forced flat.

3. Worked calculation

Four specimens were tested in each direction, face outward, after conditioning at 65% RH and 20 °C.

DIRECTION	RECOVERY ANGLES (°)	MEAN (°)
Warp	126, 130, 128, 124	(126 + 130 + 128 + 124) / 4 = 508 / 4 = **127°**
Weft	114, 118, 112, 116	(114 + 118 + 112 + 116) / 4 = 460 / 4 = **115°**

$$\text{Total crease recovery} = 127° + 115° = 242°$$

$$\text{Percentage recovery} = 242 / 360 \times 100 = 67.2\%$$

$$\text{Crease retention} = 360° - 242° = 118°$$

The individual readings deserve a glance before the means are trusted: the warp set spans only 124–130° and the weft set 112–118°, a spread of 6° in each case — tight, consistent testing with no outlier to investigate. The warp recovers 12° better than the weft (127° against 115°), the usual pattern for a woven fabric, reflecting the stronger, more closely set warp yarns. A total of 242° out of a possible 360° places the fabric in the moderate band: it will wrinkle visibly in wear and need pressing or an easy-care finish for a smart garment, but it is far from the limp, crease-holding behaviour of an unfinished linen or a soft viscose. For comparison in discussion, well-finished easy-care cotton commonly exceeds 270–300° total, wool suiting sits high on fibre recovery alone, and untreated cotton poplin often falls near 200–220° — the measured 242° is consistent with a lightly finished cotton-rich fabric.

RESULT

Warp recovery 127°, weft recovery 115°, total recovery 242° (67.2% of the full 360°), crease retention 118°. Warp exceeds weft by 12° in both mean and behaviour. Verdict: moderate crease recovery — acceptable for casual wear; an easy-care cross-linking finish would be required for a non-iron claim.

4. Precautions

- Condition specimens fully and test in the standard atmosphere (65% RH, 20 °C). A dry cotton specimen and a damp one give different angles on the same fabric, and the difference is moisture, not quality.
- Fold exactly in half and centre the fold between the plates every time. An off-centre fold gives unequal arms, an uneven crease and an angle that cannot be compared with the rest of the set.
- Time both periods strictly — 5 minutes under load, 5 minutes recovering. Reading early flatters poor fabrics disproportionately, because slow recovery is exactly what distinguishes them.
- Handle the folded specimen with tweezers by the clamped arm only. Touching, steadying or breathing the fold open during transfer adds unmeasured recovery work.
- Keep the free arm clear of the dial and all surfaces while it recovers, and read the scale at eye level to avoid parallax error.
- Cut specimens away from selvedges, existing wrinkles and handling creases, and never test a specimen that was accidentally folded before its turn — every fold has a memory and the test measures the last one imposed under standard conditions.

Practice (viva preparation)

1. Define crease recovery and crease retention, and state the maximum possible total recovery in the Shirley test.
2. Warp angles are 126, 130, 128 and 124°. Find the mean warp recovery angle.
3. Using warp 127° and weft 115°, find total recovery, percentage recovery and crease retention.
4. Why does cotton recover poorly from creasing, and how does an easy-care resin finish improve it?
5. Why are creasing time, recovery time and the standard atmosphere fixed so strictly in ISO 2313 work?
6. Why is warp recovery usually higher than weft recovery in a woven fabric?
7. State one limitation of the single-fold recovery angle as a predictor of garment appearance.

Answer key

1. Crease recovery is the angle to which a sharply folded, standard-creased specimen opens during a standard free-recovery period — the measure of how far elastic return overcomes plastic set. Crease retention is the complementary quantity, the angle of fold that remains: retention = 360° − total recovery. Since each direction can recover at most 180° (lying flat again), total recovery can reach at most 180° + 180° = 360°.
2. Mean = (126 + 130 + 128 + 124) / 4 = 508 / 4 = 127°.

3. Total = 127 + 115 = 242°. Percentage recovery = 242 / 360 × 100 = 67.2%. Crease retention = 360 − 242 = 118°.
4. In cellulose, shape is held by hydrogen bonds between chains. Bending stresses the fold zone; chains slip, bonds break and new bonds form in the folded position, locking in the crease — water accelerates the process, which is why cotton creases worst when damp. A cross-linking resin finish forms covalent bridges between chains inside the fibre, so the chains cannot slip past each other and re-bond in the folded state; the fibre is forced to spring back elastically. The gain in recovery is paid for with some loss of tensile and tear strength.
5. Because recovery is time- and moisture-dependent. The angle grows rapidly at first and then slowly, so the reading depends on when it is taken; the set imposed depends on how long the load acts; and regain changes fibre stiffness and hydrogen-bond behaviour. Fixing load, both times and the atmosphere (65% RH, 20 °C) is what makes an angle measured today comparable with one measured in another laboratory.
6. Warp yarns are typically stronger, finer or more highly twisted, are held at higher density, and are sized and tensioned for weaving — a stiffer, more resilient yarn system that stores more elastic energy in bending and springs back further. Weft yarns are usually softer, bulkier and less twisted to give cover and handle, and they yield more plastically at the fold.
7. The test bends a strip once, in one direction, under ideal conditions. A garment wrinkles in many directions at once, under body heat and moisture, and is judged by the overall pattern of sharp and soft wrinkles. A fabric can show a fair single-fold angle yet look badly rumpled in wear (or vice versa). For buying decisions the angle is therefore supplemented by replica-rating of washed or worn panels.

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